

Plant Growth & Development

Plant Development and Fundamental Concepts

- **Development Definition:** Development is the comprehensive term encompassing all changes an organism undergoes during its life cycle, from seed germination through senescence.
- **Constituent Processes:** Development is the sum of two fundamental processes: growth and differentiation.
- **Origin:** All cells in a plant are descendants of the zygote (fertilized egg).
- **Sequence of Events:** The development of a mature plant from a zygote involves a complex body organization and follows a precise and highly ordered succession of events, eventually leading to death.
- **Initial Step:** The first step in plant growth is seed germination, which occurs only when favorable environmental conditions exist. In the absence of favorable conditions, seeds enter a period of suspended growth or rest, resuming metabolic activities and growth when conditions return.

II. Plant Growth:-

1. Definition and Characteristics

- **Definition:** Growth is defined as an irreversible permanent increase in size of an organ, its parts, or even an individual cell.
- **Metabolism:** Growth is typically accompanied by both anabolic and catabolic metabolic processes that occur at the

expense of energy.

- **Indeterminate Nature:** Plant growth is unique because plants retain the capacity for unlimited growth throughout their life. This ability is due to the presence of meristems in certain locations.

- The cells of meristems can divide and self-perpetuate, but the resulting product cells soon lose the capacity to divide and form the plant body. This is also known as open growth.

- **Elongation Growth:** Root and shoot apical meristems, sometimes along with intercalary meristem, contribute to the elongation growth of plant axes.

- **Secondary Growth:** In dicotyledonous plants and gymnosperms, lateral meristems (vascular cambium and cork-cambium) appear later in life, causing an increase in the girth of the organs where they are active.

2. Measurement of Growth

- **Cellular Basis:** At the cellular level, growth is fundamentally a consequence of the increase in the amount of protoplasm.

- **Parameters Used:** Since direct measurement of protoplasm increase is difficult, growth is measured using various proportional parameters:

- Increase in fresh weight, dry weight, length, area, volume, and cell number.

- Growth can be expressed as an increase in cell number (e.g., a maize root apical meristem producing over 17,500 new

cells per hour) or an increase in cell size (e.g., watermelon cells increasing in size up to 350,000 times).

3. Phases of Growth

The period of growth is generally divided into three phases, which can be understood by examining the root tips:

1. Meristematic Phase: Represented by the constantly dividing cells at the root and shoot apices. These cells are rich in protoplasm, have large conspicuous nuclei, and possess thin, cellulosic primary cell walls with abundant plasmodesmatal connections.

2. Elongation Phase: Located proximal (just next, away from the tip) to the meristematic zone. Characteristics of this phase include increased vacuolation, cell enlargement, and new cell wall deposition.

3. Maturation Phase: Located more proximal to the phase of elongation. Cells in this zone attain their maximal size and undergo wall thickening and protoplasmic modifications.

4. Growth Rates

The increased growth per unit time is termed the growth rate.

- Arithmetic Growth: Only one daughter cell continues to divide after mitotic cell division, while the other differentiates and matures.

- A plot of the length of the organ against time yields a linear curve.

- Expressed as: .
- Geometric Growth: Both progeny cells following mitotic division retain the ability to divide and continue to do so.
 - Growth is characterized by an initial slow phase (lag phase), followed by a rapid increase at an exponential rate (log or exponential phase).
 - With limited nutrient supply, growth eventually slows down, leading to a stationary phase.
 - A plot of growth parameter against time yields a typical sigmoid or S-curve, which is characteristic of living organisms (including plant cells, tissues, and organs) growing in a natural environment.
 - Expressed as: . Here, 'r' is the relative growth rate and is a measure of the plant's ability to produce new material, referred to as the efficiency index.
- Quantitative Comparison:
 - Absolute Growth Rate (AGR): The measurement and comparison of the total growth per unit time.
 - Relative Growth Rate (RGR): The growth of the given system per unit time expressed on a common basis (e.g., per unit initial parameter).

5. Conditions Required for Growth

Essential conditions include:

- Water: Crucial for cell enlargement and turgidity, which aids in extension growth. Water also serves as the medium

for necessary enzymatic activities.

- **Oxygen:** Required for releasing the metabolic energy essential for growth activities.
- **Nutrients:** Macro and micro essential elements are needed for protoplasm synthesis and act as a source of energy.
- **Temperature:** Plants require an optimum temperature range for best growth; deviations can be detrimental.
- **Environmental Signals:** Light and gravity also affect certain phases or stages of growth.

III. Differentiation, Dedifferentiation, and Redifferentiation

- **Differentiation:** The process where cells derived from meristems (root apical, shoot apical, and cambium) mature to perform specific functions.
 - This involves structural changes in both the cell walls and protoplasm.
 - Example: Tracheary elements lose their protoplasm and develop strong, elastic, lignocellulosic secondary cell walls to carry water long distances.
- **Open Differentiation:** Differentiation in plants is considered "open" because cells/tissues arising from the same meristem possess different structures at maturity.
 - The final structure of a cell/tissue is determined by its location within the organ. For example, cells positioned away from the root apical meristem differentiate as root-cap cells, while those pushed to the periphery mature as epidermis.

- **Dedifferentiation:** The phenomenon where living, differentiated cells that have lost the capacity to divide regain the capacity for division under certain conditions.

- Example: Formation of the interfascicular cambium and cork cambium from fully differentiated parenchyma cells.

- **Redifferentiation:** Cells produced by the dividing dedifferentiated tissues subsequently lose the capacity to divide again but mature to perform specific functions.

IV. Development and Plasticity

- **Plasticity:** This is the ability of plants to follow different pathways in response to the environment or phases of life, resulting in the formation of different kinds of structures.

- **Heterophylly:** An example of plasticity.

- In plants like cotton, coriander, and larkspur, the juvenile leaves have a different shape from the mature leaves (response to phase of life).

- In buttercup, the difference in leaf shape when produced in air versus water also demonstrates heterophylly (response to environment).

V. Plant Growth Regulators (PGRs)

PGRs are intercellular intrinsic factors—small, simple molecules of diverse chemical composition—that control differentiation and developmental events. They are also referred to as plant growth substances, plant hormones, or phytohormones.

1. Functional Groups:-

- Plant Growth Promoters: PGRs involved in growth-promoting activities like cell division, cell enlargement, pattern formation, flowering, fruiting, and seed formation. (e.g., Auxins, Gibberellins, Cytokinins).
- Growth Inhibitors: PGRs involved in responses to biotic and abiotic stresses, and growth-inhibiting activities like dormancy and abscission. (e.g., Abscissic acid).
- Ethylene: The gaseous PGR, ethylene, largely acts as an inhibitor but may fit into either group.

2. Specific Physiological Effects

• Auxins:

- Examples: Indole-3-acetic acid (IAA) and Indole Butyric Acid (IBA) are natural; NAA and 2, 4-D are synthetic.
- Synthesis: Generally produced by the growing apices of stems and roots.
- Functions: Induce parthenocarpy (e.g., in tomatoes), control xylem differentiation, and help in cell division. 2, 4-D is widely used as a herbicide to kill dicotyledonous weeds without affecting mature monocotyledonous plants.

• Gibberellins (GAs):

- Nature: Promotory PGRs; acidic in nature. (Gibberellic acid) is the most intensively studied form.
- Functions: Cause an increase in the length of the axis

(used to increase the length of grapes stalks). They delay senescence, extending the market period for fruits.

- Applications: speeds up the malting process in the brewing industry. Spraying sugarcane increases stem length, raising yields. They promote bolting (internode elongation before flowering) in plants like beet and cabbages.

• Cytokinins:

- Nature: Have specific effects on cytokinesis. Kinetin was discovered in autoclaved herring sperm DNA, but natural forms like zeatin are found in corn kernels and coconut milk.

- Synthesis: Synthesized in regions of rapid cell division (e.g., root apices, young fruits).

- Functions: Promote new leaf and chloroplast production, lateral shoot growth, and adventitious shoot formation. They help overcome apical dominance. They delay leaf senescence by promoting nutrient mobilization.

• Ethylene CH_4 :-

- Nature: Simple gaseous PGR, synthesized in large amounts by senescing and ripening tissues.

- Effects: Promotes senescence and abscission of plant organs.

- Ripening: Highly effective in fruit ripening, enhancing the respiration rate—a phenomenon called respiratory climactic.

- Other Functions: Breaks seed and bud dormancy and

and initiates germination. It promotes rapid internode/petiole elongation in deep water rice plants, and promotes root growth and root hair formation, increasing absorption surface.

• **Abscissic Acid (ABA):**

- **Role:** Acts as a general plant growth inhibitor and inhibitor of plant metabolism. It was previously known as inhibitor-B, abscission II, and dormin.
- **Stress Hormone:** ABA is called the stress hormone because it increases the tolerance of plants to various stresses and stimulates the closure of stomata.
- **Dormancy:** It inhibits seed germination. It is crucial for seed development, maturation, and inducing dormancy, helping seeds withstand desiccation and unfavorable factors.
- **Antagonism:** In most situations, ABA acts as an antagonist to Gibberellins (GAs)